

International Islamic University Chittagong
Department of Computer Science and Engineering
BSc in CSE Semester Final Examination, Autumn 2025
Course Code: CSE-3521 Course Title: Computer Architecture
Total Marks: 50 Time: 2 hours 30 Minutes

Figures in the right margin indicate full marks, course outcomes, and cognitive domains.

	Group A	M	C	D
1.a)	Analyze the variation of Cycles Per Instruction (CPI) in single-cycle and multi-cycle processor architectures, and explain how instruction complexity influences CPI using a suitable instruction example.	5	3	4
	OR			
1.a)	By describing the core operations of MIPS, draw the single cycle datapath for sub \$t3, \$s2, \$s3 explaining its data flow.	5	3	4
1.b)	Demonstrate the single-cycle datapath operations to execute a Load Word (LW) instruction, and illustrate the process using a suitable instruction-level example showing the required control actions.	5	3	2
	OR			
1.b)	Explain the logic elements for multicycle datapath. Show and define how the elements handle edge-triggered clocking.	5	3	2
2.a)	Explain how the control unit generates signals for different instruction types in a multicycle MIPS datapath.	5	3	2
2.b)	In a single-cycle MIPS datapath, a 16-bit immediate value is sign-extended to 32 bits before ALU operation. Given the immediate value: 1011 0000 0000 1101. Find the 32-bit sign-extended value and the decimal value of the sign-extended number.	5	2	3
	Group-B			
3.a)	Given a sequence of dependent instructions, analyze how pipeline stalls are introduced and determine how reordering instructions can minimize performance degradation.	5	3	4
3.b)	Analyze how control signals are generated and propagated through the pipeline datapath, and explain how incorrect control handling can lead to hazards	5	3	4
4.a)	Describe the following with example: a) Multi-level Cache, b) 3Cs model.	5	2	2
4.b)	A cache consists of 64 blocks, each having a block size of 16 bytes. Determine the cache block number to which the byte address 1200 is mapped.	5	2	3
	OR			
4.a)	A program frequently accesses a few recent variables and iterates over large arrays. Predict which parts benefit from temporal and spatial locality, and justify your reasoning.	5	2	2
4.b)	Explain how a direct-mapped cache uses memory block addresses, tag, bits, and valid bits to map a memory block to a cache block.	5	2	3
5.a)	Analyze how the high cost of a page fault influences the design decisions of virtual memory systems. How can you solve it?	5	2	4
5.b)	Analyze shortly on rotation latency, RAID, hot swapping, standby spares, DMA	5	2	4

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Group-A			M	CO									
1. a)	Suppose you are given an instruction add \$s1,\$t3,\$t4. Describe the complete operation of the datapath using the instruction with a figure.		5	3									
b)	Design a multi-cycle data path for handling basic instruction of MIPS.		5	3									
2. a)	How does the execution of different instruction types differ in multi-cycle datapath. Show the steps.		5	3									
b)	An instruction set consists of the following instruction mix: 35% ALU instructions; 20% Load instructions; 15% Store instructions; 25% Branch instructions; 5% Jump instructions 1. Calculate the overall CPI for this instruction mix in a variable-cycle implementation. 2. If the processor is redesigned with a fixed-cycle, where every instruction takes 3 cycles, what is the new CPI? 3. Compare the CPI in both cases and discuss the trade-offs between using a variable-length cycle design versus a fixed-length cycle design.		5	2									
OR													
2. a)	What changes are required for the multiplexer in multicycle Datapath?		5	3									
b)	A multicycle CPU has a clock time of 250 ps and the following instruction mix: <table><tr><th>Instruction</th><th>%</th><th>Cycles</th></tr><tr><td>R-type</td><td>60</td><td>4</td></tr><tr><td>lw</td><td>40</td><td>5</td></tr></table> The program has 100,000 instructions. Q: What is the total CPU time?	Instruction	%	Cycles	R-type	60	4	lw	40	5		5	2
Instruction	%	Cycles											
R-type	60	4											
lw	40	5											
Group-B													
3. a)	Explain the concept of pipeline stalls caused by data hazards. Propose two techniques to resolve or minimize data hazards.		5	3									
b)	Draw and explain the pipeline stages involved in executing a store instruction in a pipelined architecture. Discuss how hazards are handled during execution.		5	3									
OR													
3. a)	Compare and contrast pipelined and single-cycle instruction execution. Use timing diagrams and MIPS instruction examples to support your explanation. Include assumptions about stage delays and overall speedup.		5	3									
b)	Design a pipeline schedule for the following MIPS instructions to minimize hazards: lw \$t1, 0(\$t2) add \$t3, \$t1, \$t4 sub \$t5, \$t3, \$t6 a) Identify the types of data hazards present. b) Suggest how forwarding or stalling could resolve them. c) Reorder the instructions (if possible) to minimize stalls.		5	3									
4. a)	Discuss the working of a direct-mapped cache. Include the addressing scheme and explain how hits and misses are determined.		5	2									
b)	What is virtual memory? How does the page table help translate virtual addresses into physical addresses?		5	2									
5. a)	Describe how Translation Lookaside Buffer (TLB) improves memory access time. What happens on a TLB miss?		5	2									
b)	What is handshaking in asynchronous communication? Describe its steps with an example of a read request from an I/O device.		5	2									

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Answer all Five of the following questions. The figures in the right-hand margin indicate full marks.

Group - A

	M	C	D
1.a) Describe the edge triggered clocking methodology.	4	1	
b) Suppose you are given an instruction add \$s1,\$t3,\$t4. Describe the complete operation of the datapath using the instruction with a figure.	4	1	
c) Differentiate multicycle over single cycle datapath?	2	2	
2.a) Points out different steps and benefits of fetch execute cycle	3	5	
b) Design and describe the working procedure of jump instruction in single-cycle data path.	7	5	
OR			
2.a) Write down the basic characteristics of RISC and CISC architectures .	4	1	
b) What is exception? How exception are handled?	4	1	
c) What temporary registers are used in multicycle datapath?	2	2	

Group - B

3 a) Does the use of pipelining reduce, increase or leave unchanged the execution time of a single machine instruction?	3	3	
b) How can you resolve data hazard -explain with figure.	5	1	
c) Dynamic pipelining is more complicated than the traditional or static pipelining – why?	2	2	
4.a) How does spatial locality be helpful in handling cache misses? Explain with example.	3	3	
b) Explain the working principle of direct mapped cache with figure. How many total bits are needed for a direct mapped cache with 64KB data in 1 word blocks?	7	3	
5 a) Explain the working principle of TLB with figures.	5	3	
b) What is the difference between synchronous bus and asynchronous bus? We want to compare the maximum bandwidth for a sync and async bus. The sync bus has a clock cycle time of 50ns and each bus transmission takes 1 cycle. The async bus requires 40ns per handshake. Each bus has 32 data lines. What is the bandwidth when performing one word reads from a 200ns memory?	5	4	
OR			
5 a) Explain cache read hit/miss.	2	1	
b) Define miss penalty, cache miss, hit rate, tag in caching.	4	1	
c) What is DMA controller? Describe the configuration of DMA controller.	4	1	

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[Answer all questions from **Group-A** and **Group-B**
 Separate answer script must be used for Group-A and Group-B.]

Group-A

- | | | |
|--|---|-----|
| 1.a) What is hardwired control? How it is different from micro-programmed control? | 2 | CO1 |
| b) Suppose you are given an instruction <code>sub \$s1,\$t3,\$t4</code> . Describe the complete operation of the datapath using the instruction with a figure. | 5 | CO1 |
| Or, | | |
| Illustrate the following instruction with three bus and two bus CPU system and explain which is better? <code>ADD R3, R2, R1</code> | | |
| c) Why a single-cycle implementation is not used? justify the answer. | 3 | CO2 |
| 2.a) Do you think that the functions of memory data register and instruction register are the same? If not then how? | 3 | CO1 |
| b) Write down the basic characteristics of RISC and CISC architectures. | 5 | CO1 |
| Or, | | |
| Design a multi-cycle data path for handling basic instruction of MIPS. | | |
| c) What changes are required for the multiplexer in multicycle datapath? | 2 | CO2 |

Group-B

- | | | |
|---|---|-----|
| 3.a) Does the use of pipelining reduce, increase or leave unchanged the execution time of a single machine instruction? | 3 | CO3 |
| b) How can you resolve data hazard -explain with figure. | 5 | CO1 |
| c) Dynamic pipelining is more complicated than the traditional or static pipelining – why? | 2 | CO2 |
| 4.a) Remark on the purpose, scope and mechanisms of the followings: i) Programmed I/O
ii) Interrupt Initiated I/O iii) Direct Memory Access. | 3 | CO4 |
| b) What is a DMA controller? Write the configuration of the DMA controller. | 5 | CO1 |
| Or | | |
| What is TLB? How can you reduce page fault? Explain logically? | | |
| c) Define miss penalty, cache miss, hit rate, tag in caching. | 2 | CO1 |

5.a) Find the Location of a memory block with address 14 in a cache with 8 blocks with different degrees of associativity

5 CO2

Or

When will the page faults occur? What is the procedure for handling the page fault?

b) What are the key factors for designing a virtual memory? What are similarities between cache and virtual memory?

3 CO1

c) Show the comparative analysis of write through scheme and write back scheme.

2 CO1

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[Answer all questions from **Group-A** and **Group-B**
 Separate answer script must be used for Group-A and Group-B.]

Group-A

- 1.a) Discuss the clocking methodology for datapath implementation. 3 CO1
 b) Suppose you are given an instruction `sub $s2,$t2,$t3`. Describe the complete operation of the datapath using the instruction with a figure. 4 CO1

Or,

Suppose you are given an instruction `lw $s1,100($s3)`. Describe the complete operation of the datapath using the instruction with a figure

- c) What are the advantages of multicycle implementation over single cycle? 3 CO1
 2.a) Design a single cycle data path for handling R-type instruction of MIPS. 4 CO1

Or,

Design a multi-cycle data path for handling basic instruction of MIPS.

- b) Describe all temporary register and multiplexors of Multicycle Datapath. 4 CO1
 c) Do you think that the functions of memory data register and instruction register are the same? If not then how? 2 CO1

Or,

Why a single-cycle implementation is not used? justify the answer.

Group-B

- 3.a) Does any Pipeline Stall arise here for the following instructions----- explain: 3 CO1
`MUL R1, R2, R3`
`SUB R3, R1, R4`
`ADD R4,R5,R6`
 b) What do you mean by pipelining? Describe advantages of pipelining architecture. 5 CO1
 c) Dynamic pipelining is more complicated than the traditional or static pipelining – why? 2 CC
 a) Write short note on structural hazard. Or, Write short note on memory hierarchy. 2 CC
 b) Explain interrupt and its classes with necessary diagram. Or, What is TLB? Discuss the operation of paging and TLB using flow diagram? 4 CC
 c) What is DMA controller? Describe the configuration of DMA controller. 4 CC
 a) Discuss the clocking methodology with relevant diagram 6 CC
 b) Explain the handshaking protocol with diagram. Or, Explain the direct mapped cache system with figure. 4 CC

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[Answer all the questions; Separate answer script must be used for Group-A and Group-B.]

Group-A

- | | | |
|--|---|-----|
| 1.a) What are the basic elements in an MIPS data path implementation? | 3 | CO1 |
| b) Suppose you are given an instruction <code>sub \$s2,\$t2,\$t3</code> . Describe the complete operation of the datapath using the instruction with a figure. | 4 | CO1 |
| c) Why a single-cycle implementation is not used? justify the answer | 3 | CO2 |
| 2.a) Design a single cycle data path for handling R-type instruction of MIPS. | 3 | CO1 |
| b) Describe all temporary register and multiplexors of Multicycle Datapath | 4 | CO2 |
| c) Do you think that the functions of memory data register and instruction register are the same? If not then how? | 3 | CO3 |
| Or, | | |
| 2. a) Are instruction and microinstruction same? If not then why? | 3 | CO1 |
| b) Illustrate the following instruction with three bus and two bus CPU system and explain which is better?
<code>ADD R3, R2, R1</code> | 4 | CO2 |
| c) Set the control words for the following microinstructions:
i) <code>SUB R1, R2, R0</code> , ii) <code>MAR ← PC</code> | 3 | CO3 |

Group-B

- | | | |
|--|---|-----|
| 3.a) Does any Pipeline Stall arise here for the following instructions, explain:
<code>MUL R1, R2, R3</code>
<code>SUB R3, R1, R4</code>
<code>ADD R4, R5, R6</code> | 4 | CO2 |
| b) Which is advantageous between sequential processing and pipeline processing, explain? | 2 | CO2 |
| c) How does data dependency effect in pipelining? | 4 | CO2 |
| 4.a) Suppose you have 6 I/O devices in your computer. An output operation will be performed to the Device 4 (could be the keyboard) in the case of shared I/O arrangement. Explain with diagram how the operation will be performed. | 5 | CO3 |
| b) Suppose you would like to bi-pass the CPU to perform a job. How a DMA can work in this regard? | 5 | CO1 |
| 5.a) What is DMA controller? Describe the working principle of DMA controller. | 3 | CO1 |
| b) What is TLB? How can you reduce page fault explain logically? | 3 | CO3 |
| c) Write short note on handshaking protocol. | 4 | CO1 |
| Or, | | |
| 5a) Write short note on memory hierarchy. | 3 | CO1 |
| b) Define <i>cache hit</i> , <i>cache miss</i> , <i>hit rate</i> , <i>tag</i> in caching. | 4 | CO1 |
| c) Suppose you would like to bi-pass the CPU to perform a job. How a DMA can work in this regard? | 3 | CO3 |